

Inhyuk Park

UNDERGRADUATE, COMPUTER SCIENCE AND ENGINEERING · MATHEMATICAL SCIENCES (DUAL MAJOR)

Seoul National University, Seoul, South Korea

☎ +82 10-4438-9335 | ✉ alex0211@snu.ac.kr | 🏠 alexander050211.github.io | 📧 alexander050211

Education

Seoul National University

Seoul, South Korea

BS IN COMPUTER SCIENCE AND ENGINEERING WITH MATHEMATICAL SCIENCES (DUAL MAJOR)

Mar 2024 - present

- GPA: 4.11 / 4.30 CSE Major GPA: 4.13 / 4.30 All A+ in math coursework
- Currently on a leave of absence.

Gyeonggi Science High School for the Gifted

Suwon, South Korea

HIGH SCHOOL DIPLOMA

Mar 2021 - Jan 2024

- GPA: 4.14 / 4.30, graduated 22 / 125
- Certificate of Excellence (Graduation Thesis in Mathematics)

Research Experience

AFD Lab, Seoul National University (Advisor: Prof. Chenglin Fan)

Seoul, South Korea

UNDERGRADUATE RESEARCH INTERN

Dec 2025 - present

- Differentially private subgraph counting; sublinear graph algorithms; proofs of proximity.
- “Differentially Private Proofs of Proximity for Bounded-Degree Graphs” — under review, ITCS 2027.
- “When Does Shuffling Help? An Algebraic Parity Split for DP Cycle Counting” — under review, AAAI 2027.

Development of Mapping Methods for Image Encryption over Elliptic Curve

GRADUATION THESIS, GYEONGGI SCIENCE HIGH SCHOOL FOR THE GIFTED

2022 - 2023

- Developed an image encryption scheme to reduce the memory used in the transmission process.
- Verified the security using image histogram analysis, NPCR, and UACI tests.
- Improved encoding time and memory usage over the Koblitiz method.

Crested Honey Buzzard Optimization Algorithm

SCIENCE-GIFTED CREATIVE RESEARCH (R&E), MINISTRY OF SCIENCE AND ICT

2023

- Meta-heuristic optimization algorithm inspired by the hunting strategy of the crested honey buzzard.
- Built separate models for exploration and exploitation; benchmarked against standard test functions.
- Improved performance over GWO and WOA on a subset of benchmark functions.

Randomness Analysis of Hash Functions with Bool Function Replacements

R&E PROJECT, GYEONGGI SCIENCE HIGH SCHOOL FOR THE GIFTED

2021

- Replaced Bool functions in SHA-256 and MD5 to improve randomness, using non-linear LFSR combinations.
- Improved randomness in 6/15 NIST SP 800-22 subtests for SHA-256 and 5/15 for MD5.

Honors and Awards

2026	Bronze Award (Nexon Foundation Chairman's Award) , Top 10 of 1,603 teams, NYPC 2026 Master Track, Nexon Youth Programming Challenge
2026	Third Place , 2026 Jane Street Electronic Trading Challenge (Seoul), Jane Street Group
2026	Selected for the Kim Jae-chul AI Class , Seoul National University, Kim Jae-chul Graduate School of AI
2025	High Honor (23rd place) , 2025 ICPC Asia Seoul/Busan Regional Contest, the ICPC Executive Director
2025	Finalist , Samsung Collegiate Programming Challenge (Section: Algorithm), the Vice Chairman and CEO of Samsung Electronics Co., Ltd.
2024	Presidential Science Scholarship , the Acting President of South Korea
2024	High Honor (19th place) , 2024 ICPC Asia Seoul Regional Contest, the ICPC Executive Director

2023	Commendation , the President of the Korean Middle School Principals Association
2023	Bronze Prize , 1st and 2nd Round of Korean Olympiad in Informatics, Olympiad in Informatics Committee Chair
2023	Encouragement Prize , Samsung Humantech Paper Awards (Section: Mathematics & Data Processing), the Vice Chairman and CEO of Samsung Electronics Co., Ltd.
2022	Bronze Prize , Fall Conference R&E Poster Presentation Competition, the President of the Korean Society of Mathematical Education (KSME)
2022	Bronze Prize , 2nd Round of Korean Olympiad in Informatics, the President of the Korean Institute of Information Scientists and Engineers (KIISE)
2022	Silver Prize , 1st Round of Korean Olympiad in Informatics, the President of KIISE
2022	Completion , International Olympiad in Informatics Winter School, the President of KIISE
2021	Paper Accepted , Fall Conference, the Korean Society for the Gifted
2021	Paper Accepted , Fall Conference, KSME
2021	Completion , International Olympiad in Informatics Summer School, the President of KIISE
2021	Bronze Prize , 1st and 2nd Round of Korean Olympiad in Informatics, the President of KIISE

Selected Projects

NEXT VISION Game AI — NYPC 2026 Master Track Finals

Seoul, South Korea

TEAM CPG, WITH JINSU KIM, SEHYEON PARK, AND JINYOUNG PARK

Aug 2026

- Main bot developer: advanced the team's primary bot through eight generations to the final submission, in a six-hour on-site contest.
- Exposed ~100 decision thresholds — defense radius, attack-launch troop count, expansion targets — as compile-time constants, so strategy variants could be built and scored by automated matches without source edits; dozens were validated inside the contest's 15-minute evaluation cycle.
- Top 10 of 1,603 entrant teams — Bronze Award, from a 23-team on-site final.
- Tools: C++.

TCG Opal SSD Protocol Compliance Verification — Basics of Deep Learning Term Project

Seoul, South Korea

TEAM OF 4 WITH JINYOUNG PARK, SEUNGJAE BANG, AND DONGHYEON LEE

Jun 2026

- 1st of 34 teams (90.7 private leaderboard score; 91.5% public accuracy). Task: decide whether the final response in a TCG Opal SSD command-response trajectory is protocol-compliant.
- Built the AI-agent development loop that surfaced missing protocol rules from the TCG Core and Opal specifications, and contributed to the 8,000-case generated test set.
- The team's deterministic state-machine verifier beat every learned baseline evaluated: LoRA fine-tuning 81.5%, Graph RAG 77.5%, hybrid 88.5%.
- Tools: Python.

Magic Show — Jane Street ETC 2026 Trading Bot

Seoul, South Korea

TEAM OF 3 WITH DONGHYEON LEE AND DANIEL CHOI

May 2026

- Market-making and arbitrage bot built in 4 days for Jane Street's Electronic Trading Challenge (Seoul); placed 3rd.
- Cross-asset risk model with liquidity-weighted (microprice) fair-value; ETF basket arbitrage with conversion-fee handling.
- Python over synchronous TCP/JSON.

Internships

Kookmin University — Department of Financial Information Security

Seoul, South Korea

LABORATORY INTERNSHIP

Jul 2022

- Research Report: "Elliptic Curve Cryptography for Image Encryption and Key Exchange."

Kookmin University — Department of Financial Information Security

Seoul, South Korea

LABORATORY INTERNSHIP

Jan 2022

- Research Report: "TRNG using Raspberry Pi."

Skills

PROGRAMMING LANGUAGES

C++, Python, Go, Java, JavaScript / TypeScript

DEVELOPMENT

Front-end web: HTML, CSS, TypeScript, React, Next.js Cross-platform: React Native, Expo

PROBLEM SOLVING

Candidate Master on Codeforces (handle: alex0211)

ENGLISH PROFICIENCY

TEPS 549 (Mar 2025), TOEIC 980 (Jan 2022)